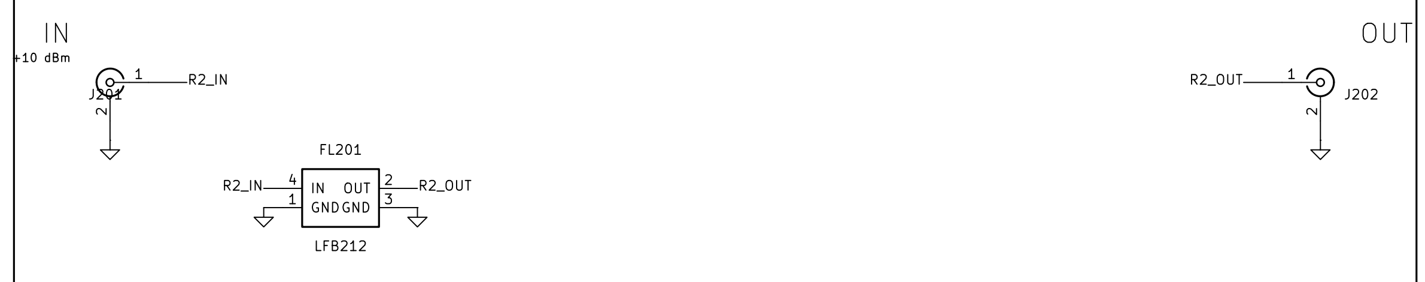
R2 2.45 GHz filter

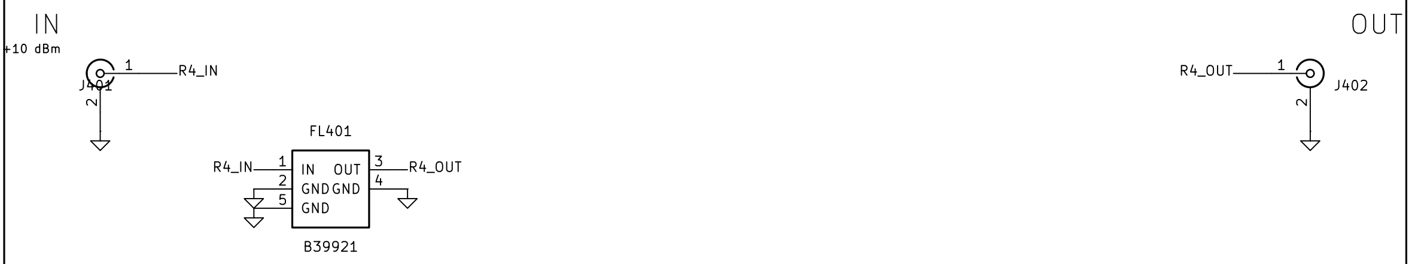
cell (2,0)



EXPECT: 1.7 dB of loss from 2.4 to 2.5 GHz, 30 dB or more below 2.2 GHz and above 2.7 GHz (the datasheet); 50 ohm in and out
Murata LFB212G45SG8A127, a 2012–size LTCC band–pass. The line to each jack is 50 ohm coplanar waveguide (0.29 mm wide, 0.25 to the ground pour) with a via fence; the jack's centre pin has no plane under it (a 2.2 mm void).
Sweep it with the synthesizer R10 or R11 into a power meter, or with a DAC harmonic comb and the ADC.

R4 915 MHz filter

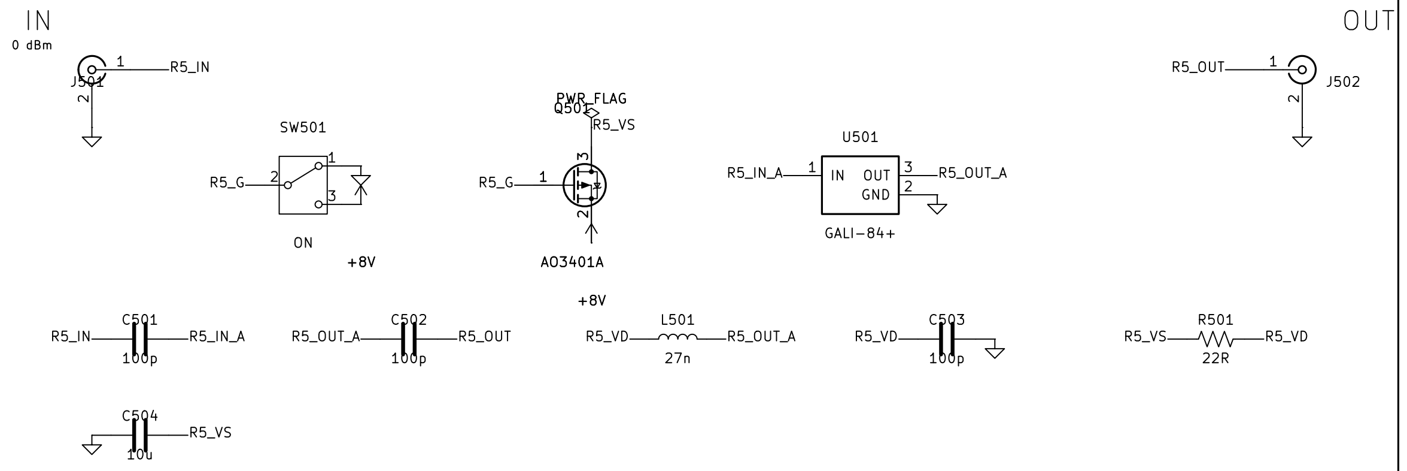
cell (3,0)



EXPECT: 1.1 dB of loss from 902 to 928 MHz, 40 dB or more 30 MHz outside the band (the datasheet); 50 ohm in and out
RF360 B39921B2672P810: a surface–acoustic–wave filter for the 915 MHz ISM band, 26 MHz wide, in a 1411 package;
its skirts are much steeper than the LTCC's. Lines, fences and launch voids as on R2.

R5 gain block, GALI–84+

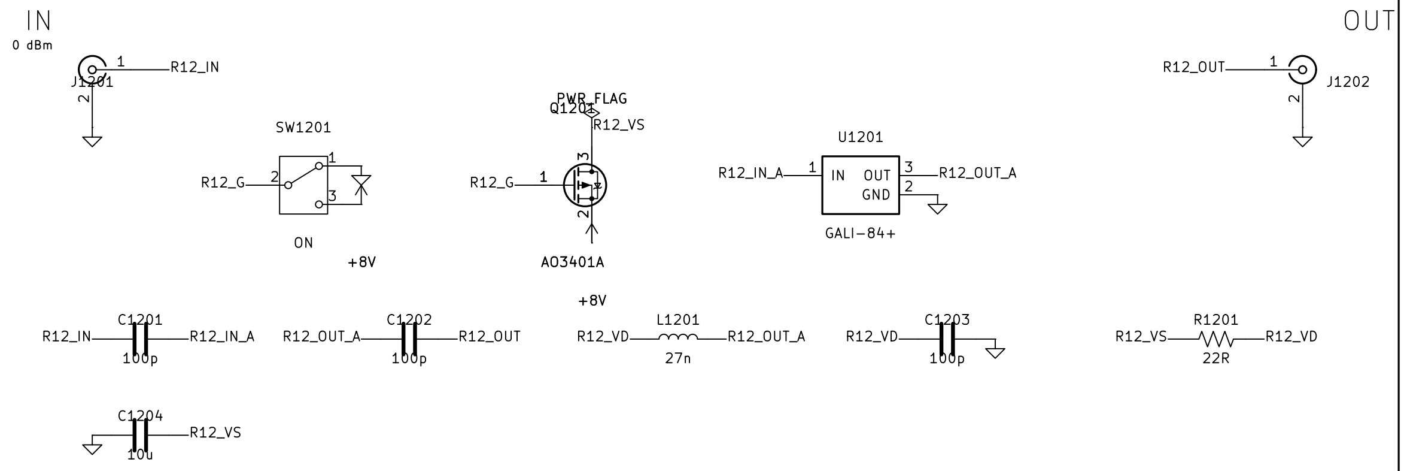
cell (0,1)



EXPECT: 20 dB of gain from 50 MHz to 2.5 GHz (datasheet: 21.9 dB at 1 GHz, 1 dB compression at +20 dBm out); 97 mA from 7.8 V
Mini–Circuits GALI–84+: a Darlington MMIC that sets its own current; the 22 ohm from the 7.8 V rail drops to its 5.8 V device voltage (0.2 W in the 2512). The 27 nH choke (415 ohm at 2.45 GHz, 155 at 915 MHz) feeds the output pin; 100 pF blocks keep the DC off the jacks. Keep the input under 0 dBm for linear gain. The slide switch and P–FET gate the tile's 7.8 V as on AMOD–A.

R12 gain block, GALI–84+

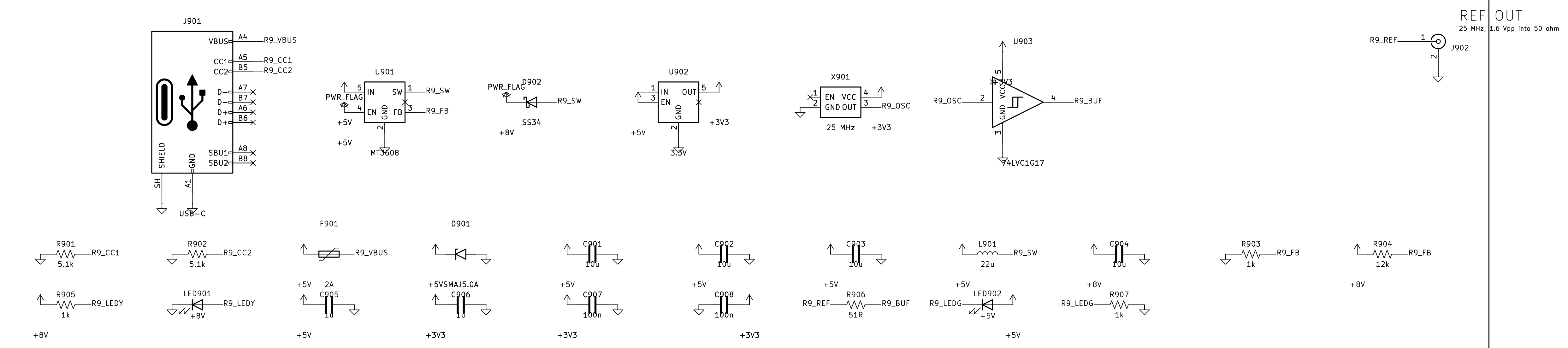
cell (1,1)



EXPECT: 20 dB of gain from 50 MHz to 2.5 GHz (datasheet: 21.9 dB at 1 GHz, 1 dB compression at +20 dBm out); 97 mA from 7.8 V
Mini–Circuits GALI–84+: a Darlington MMIC that sets its own current; the 22 ohm from the 7.8 V rail drops to its 5.8 V device voltage (0.2 W in the 2512). The 27 nH choke (415 ohm at 2.45 GHz, 155 at 915 MHz) feeds the output pin; 100 pF blocks keep the DC off the jacks. Keep the input under 0 dBm for linear gain. The slide switch and P–FET gate the tile's 7.8 V as on AMOD–A.

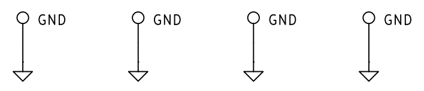
R9 power, USB–C 5 V in: +5, +7.8 V; 25 MHz reference

cells (2,1), (3,1)

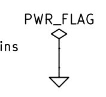


EXPECT: +5 V from the USB–C (up to 1.5 A), +7.8 V at 250 mA for the gain blocks (yellow LED), 25 MHz square wave on REF OUT and the REF25 bus
USB–C power only: 5.1 k on each CC asks any source for 5 V; a 2 A PTC, a 5 V TVS and 20 uF. The MT3608 boosts to 7.8 V (0.6 V x 13 from 12 k / 1 k) for the two GALI–84+ tiles (200 mA); the +8V rail is a strip under row 1 that their headers reach with a via. The TPS7A2033 makes 3.3 V for the TC32H4 TCXO (+–2 ppm) and the 74LVC1G17 that drives REF OUT (51 ohm: 1.6 Vpp into 50 ohm) and the REF25 line on the back to the synthesizers' jumpers (3.3 Vpp open).
The TCXO's pad 1 (enable or no connection on the drawing) floats, as CostasLink's review settled.

M3 mounting holes, one per corner



ground loops for the scope's clips, one per channel, in the board's margins



Drawn by scripts/gen_sch.py from boards/rf_mod.py; the expected responses from sim/passive.py
Every tile is one circuit with its own SMA jacks; the board's silkscreen draws each tile's wires as they
printed antennas, filters, an LNA, a 2.4 GHz front end, two mixers, two synthesizers, a gain block, a q
Board: Jason Gallicchio and Claude | open hardware, CERN–OHL–P–2.0 | github.com/gallicchio/
Sheet: /
File: rf_mod.kicad_sch
Title: RF–MOD: 915 MHz, 2.4 GHz and a stretch to 5.8 GHz
Size: A1
Date: 2026–10–25
Rev: A
KiCad E.D.A. 10.0.7
Id: 1/1